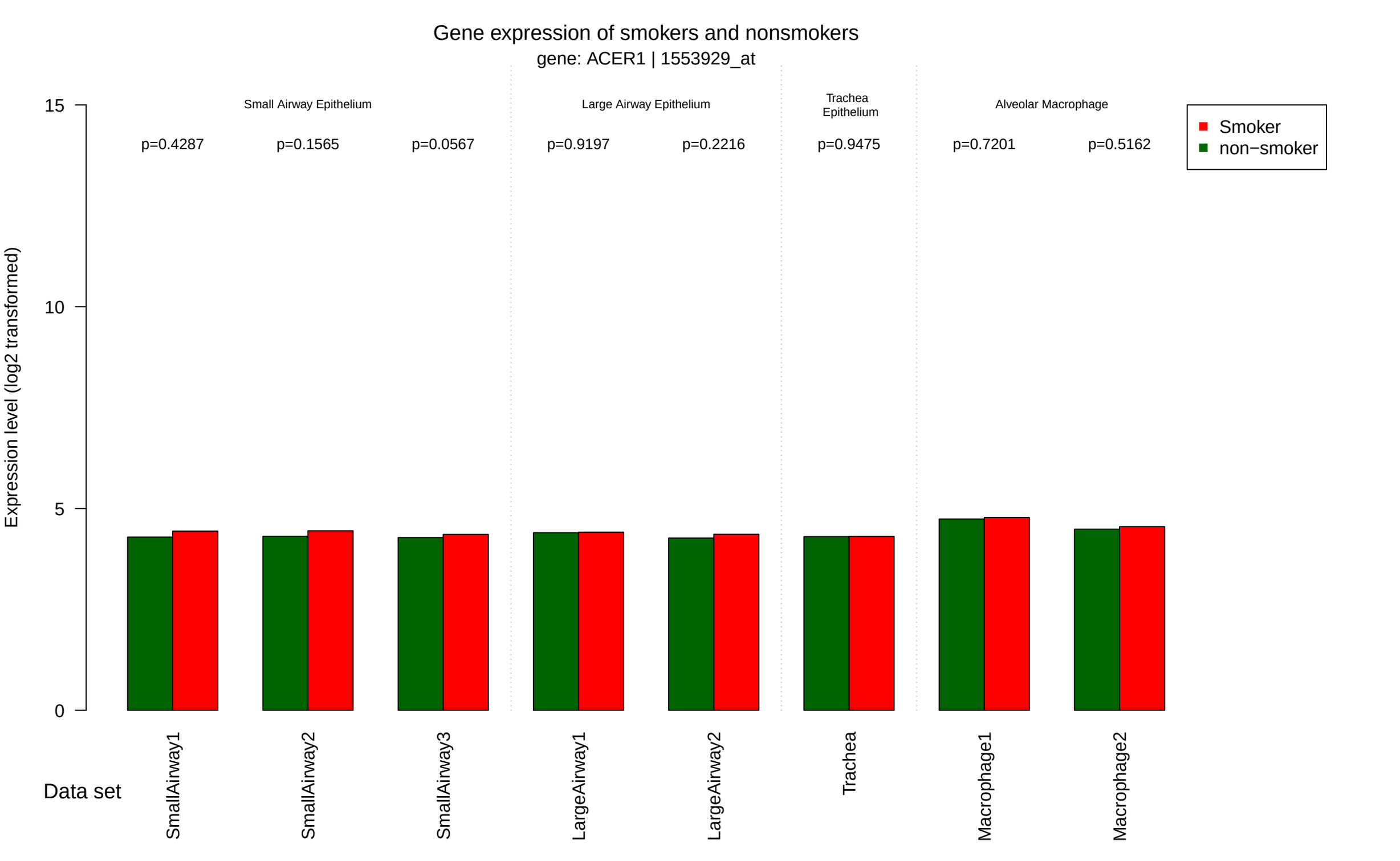
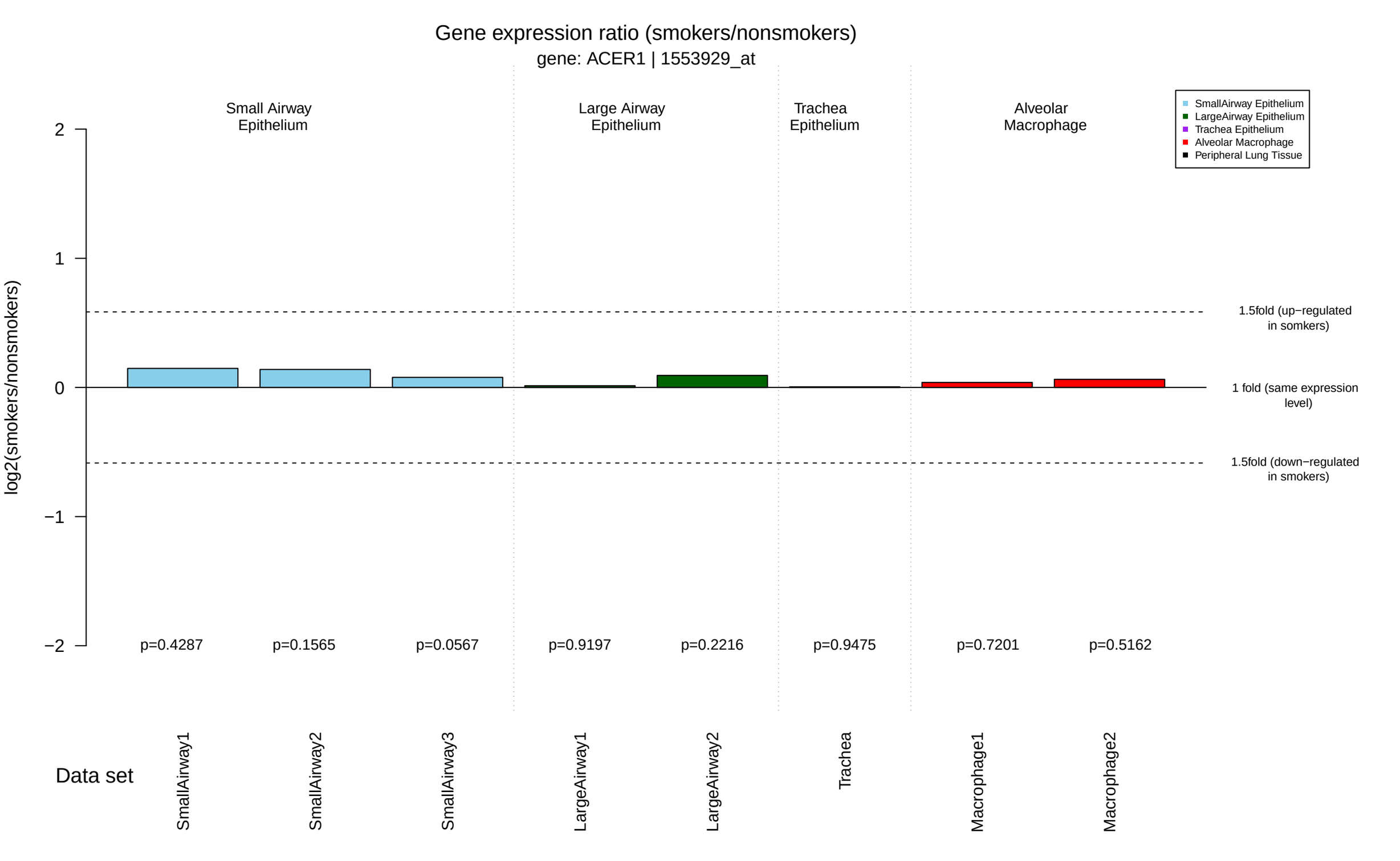
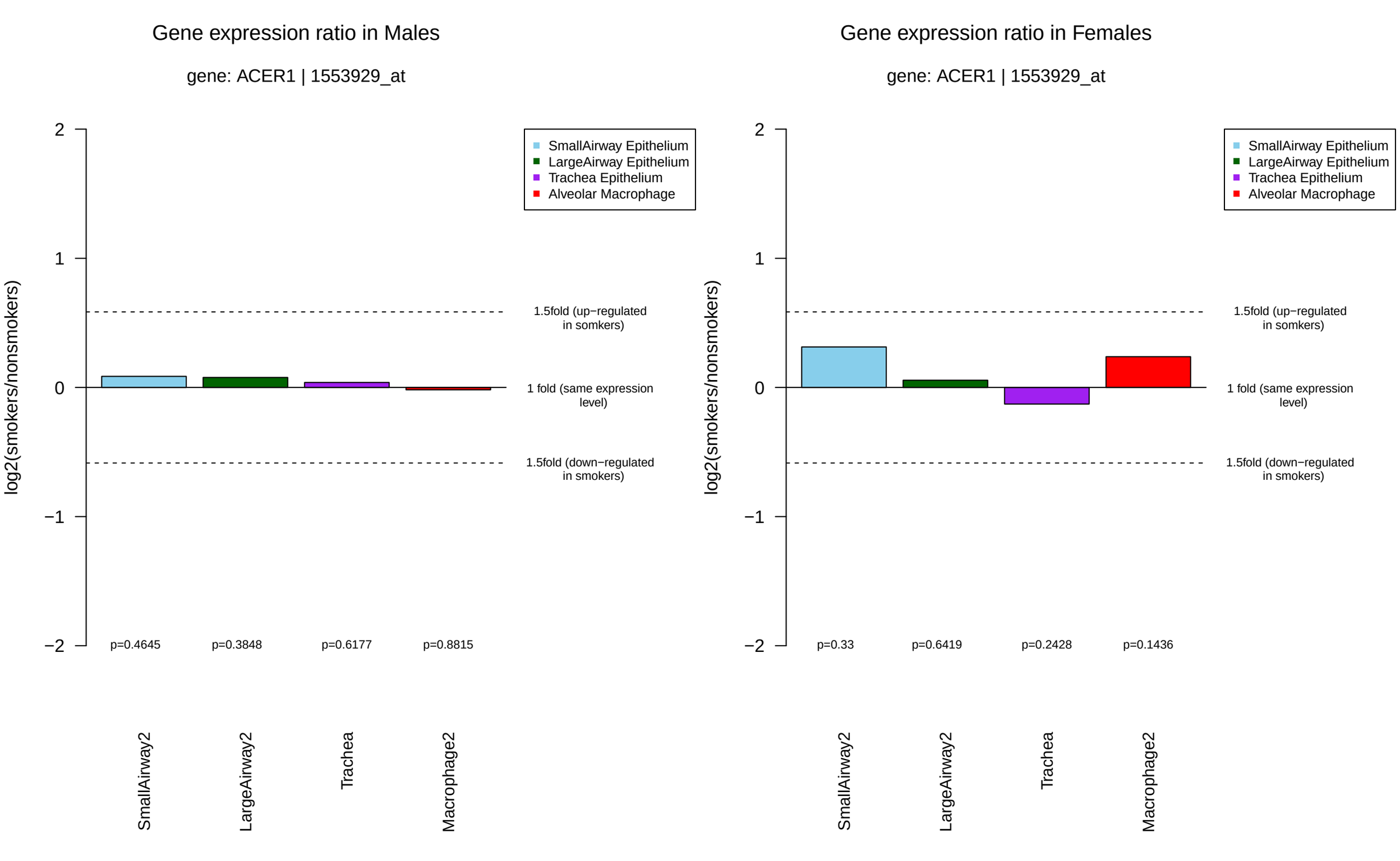
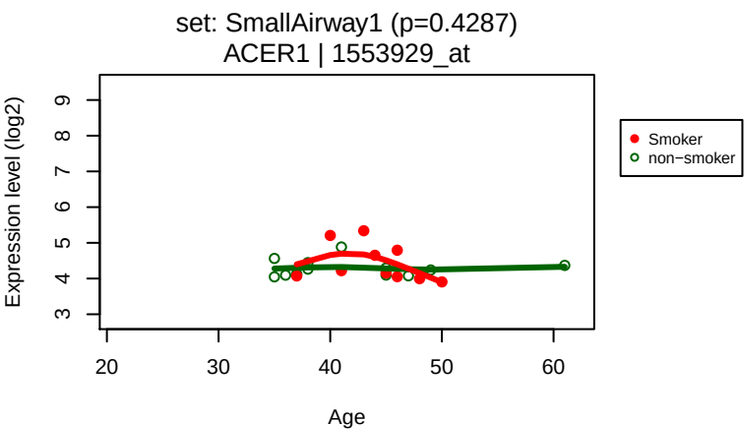
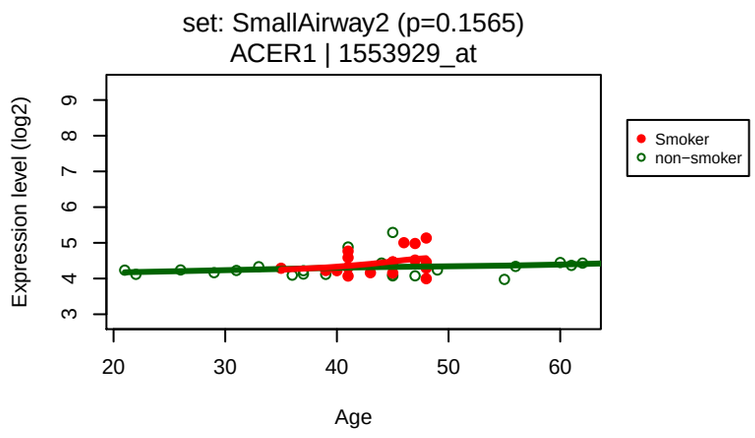
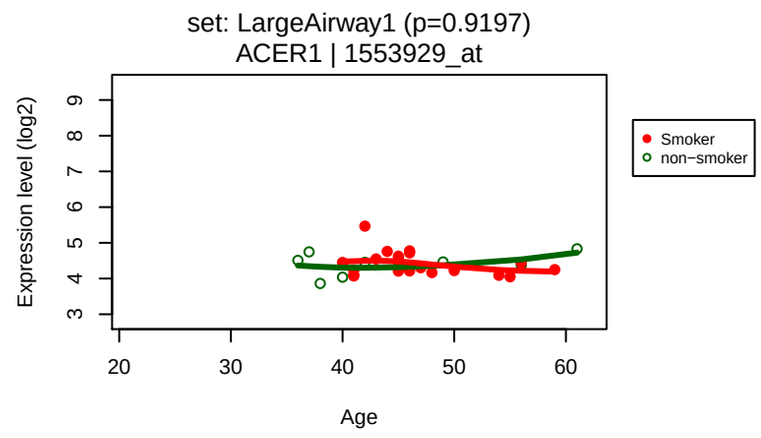

Gene expression of smokers and nonsmokers
gene: ACER1 | 1553929_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4287
p=0.1565
p=0.0567
p=0.9197
p=0.2216
p=0.9475
p=0.7201
p=0.5162
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

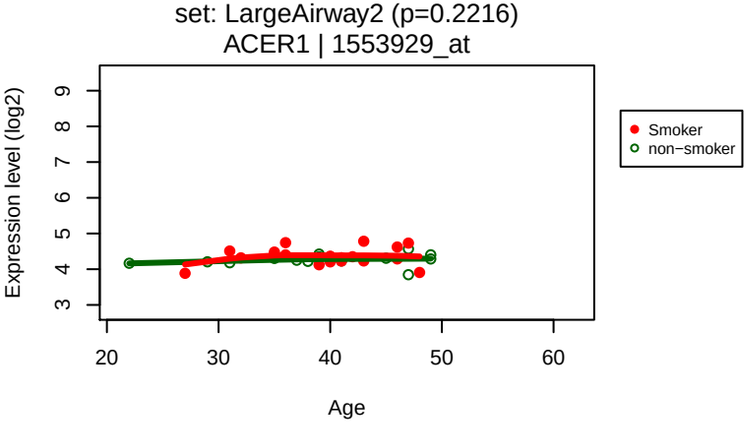

Gene expression ratio (smokers/nonsmokers)
gene: ACER1 | 1553929_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4287
p=0.1565
p=0.0567
p=0.9197
p=0.2216
p=0.9475
p=0.7201
p=0.5162
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

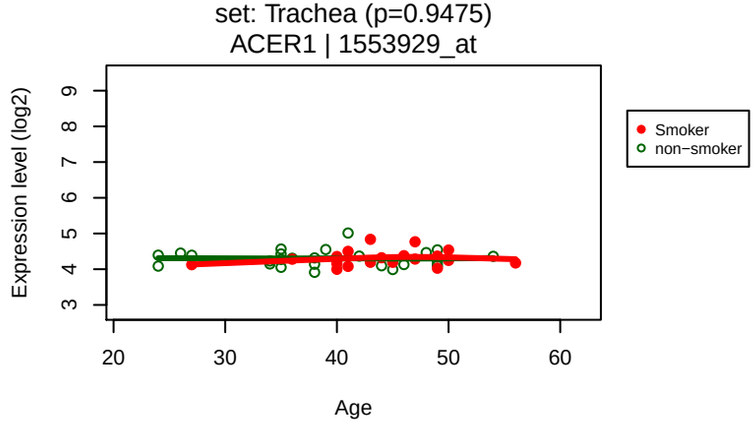

Gene expression ratio in Males
gene: ACER1 | 1553929_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4645
p=0.3848
p=0.6177
p=0.8815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACER1 | 1553929_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.33
p=0.6419
p=0.2428
p=0.1436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

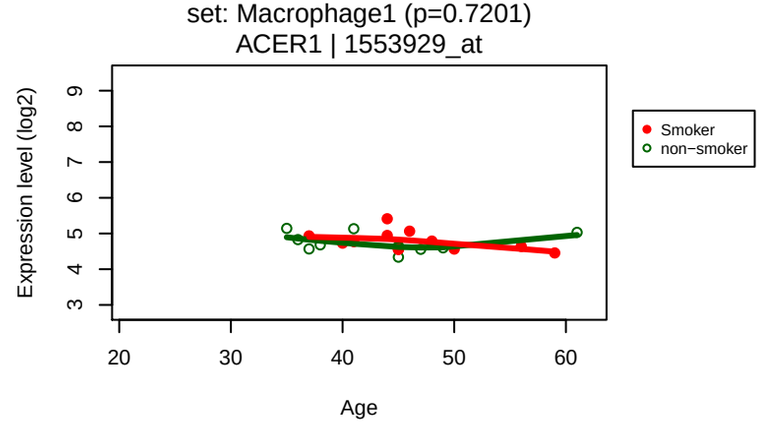

set: SmallAirway1 (p=0.4287)
ACER1 | 1553929_at
Expression level (log2)
Age
Smoker
non−smoker

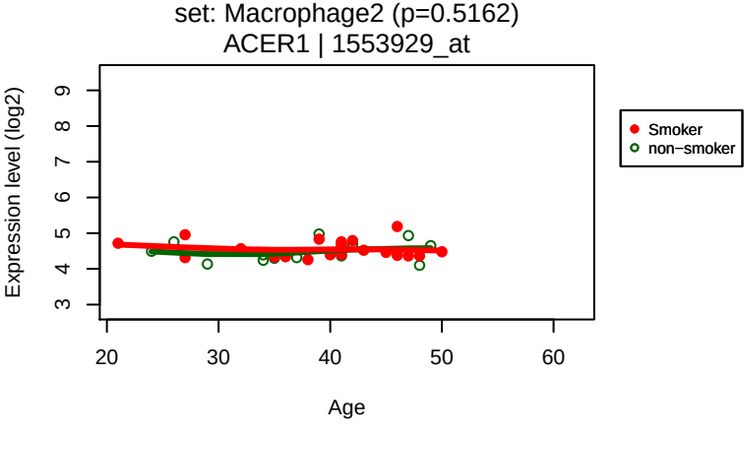

set: SmallAirway2 (p=0.1565)
ACER1 | 1553929_at
Expression level (log2)
Age
Smoker
non−smoker

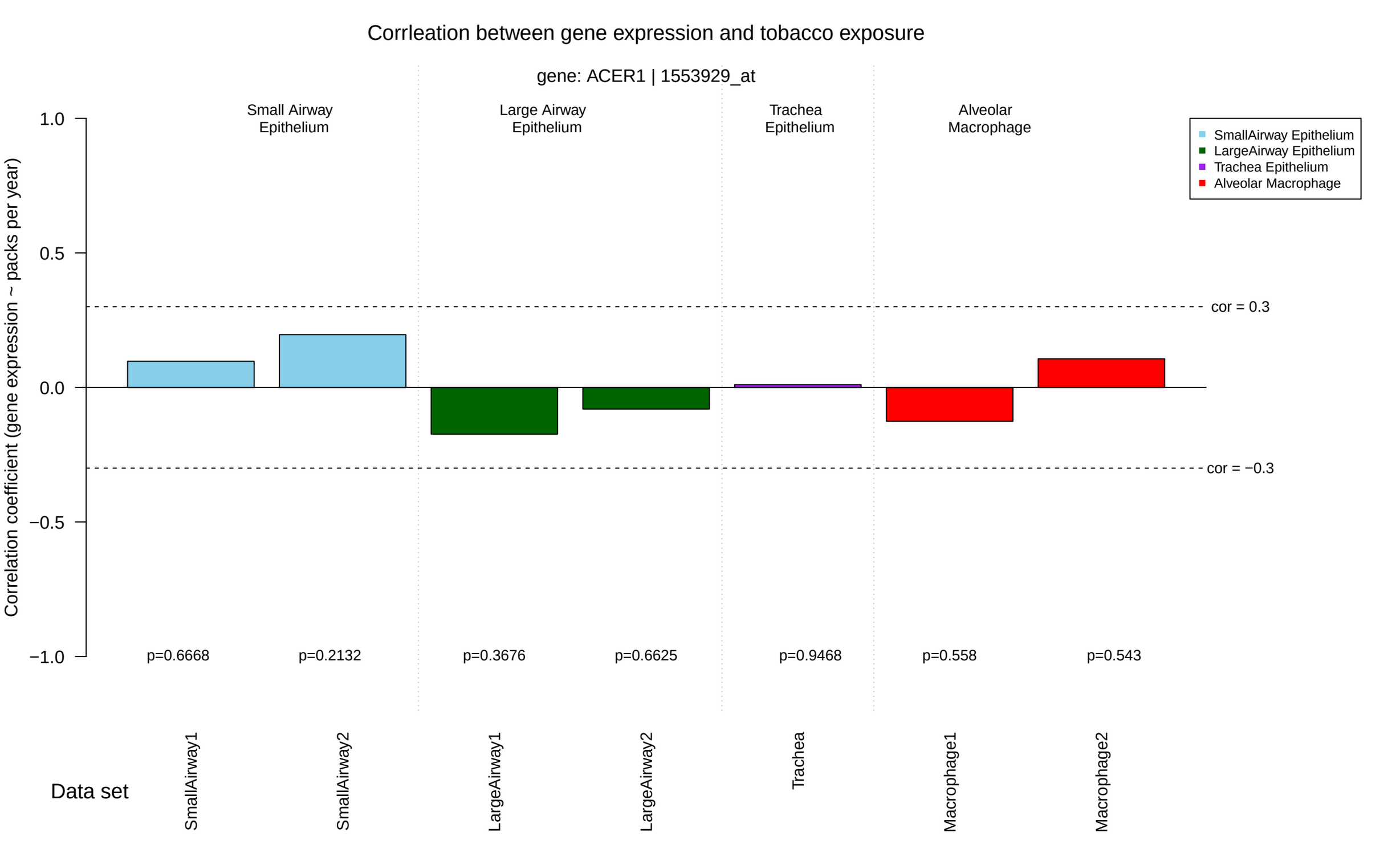

set: LargeAirway1 (p=0.9197)
ACER1 | 1553929_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2216)
ACER1 | 1553929_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9475)
ACER1 | 1553929_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7201)
ACER1 | 1553929_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5162)
ACER1 | 1553929_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACER1 | 1553929_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6668
p=0.2132
p=0.3676
p=0.6625
p=0.9468
p=0.558
p=0.543
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2